

7(a)	any three from: 1 opens (voltage-gated) Ca^{2+} channels in sarcoplasmic reticulum or calcium ions leave sarcoplasmic reticulum ; 2 calcium ions bind to troponin ; 3 troponin changes shape / tropomyosin moves ; 4 exposes binding site on actin ; 5 myosin <u>head</u>, binds to (binding) site / forms cross bridge ; plus 6 myosin <u>head</u>, tilts / AW ; 7 pulls actin / power stroke (so sarcomere shortens) ;	4
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Question	Answer	Marks									
7(b)	any two from: 1 young mice have more (muscle fibres) that are smaller (in diameter) ; ora 2 young mice have smaller range (of diameters of muscle fibres) ; ora 3 comparative data quote ; e.g. <table border="1"> <tr> <td></td><td>number of muscle fibres at mean diameter / μm</td><td>spread of diameters / μm</td></tr> <tr> <td>young mice</td><td>35 at 30</td><td>16–44</td></tr> <tr> <td>adult mice</td><td>16 at 50</td><td>20–80</td></tr> </table> plus any two from: young mice 4 fewer, (muscle) fibres / myofibrils / sarcomeres ; ora 5 less, muscle protein / actin and myosin ; ora 6 so, weaker contraction / AW ; ora		number of muscle fibres at mean diameter / μm	spread of diameters / μm	young mice	35 at 30	16–44	adult mice	16 at 50	20–80	4
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